



NATIONAL SCHOOL OF BUSINESS MANAGEMENT

BSc (Hons) Software Engineering (NSBM) - 16.2/17.1

4th Year 2nd Semester Examination

02 November 2020 , 5.30 PM to 8.30 PM

SE404.3– Agent Based Systems

Instructions to Candidates

- 1) **Answer all questions**
- 2) The total time allocated for the examination is three (03) hours.
- 3) Total number of pages - Five (05)
- 4) If a page or a part of this question paper is not printed, please inform the Supervisor immediately.
- 5) Write your index number in all pages of the answer script.
- 6) Staple all answer sheets at the end of the examination.

Question -1 (Insects moving in a grass field)**[15 marks]**

- 1) Explain the basic model rules used in modeling insects moving in a grass field (8 marks)
- 2) Describe a way of effectively displaying the growth of grass? Note: code is not required. (3 marks)
- 3) Imagine that group of bats are now feasting on the insects. Explain how you will model this scenario and how you will monitor the current population of each species? (4 marks)

Question -2 (Spreading of a Virus)**[25 marks]**

- 1) Explain the rules used in modeling a spread of virus (8 marks)
- 2) In a spread of disease model, contact results in infection, but in most real-world diseases every time people come into contact they do not necessarily spread disease. Moreover, some people are more resistant to disease than others. What would be the best way to have the individuals not spread the virus every time they come in contact, and to have some individuals who are less likely to get the infection? (5 marks)
- 3) What other variables might affect disease spread and how might you include them in the model? (6 marks)
- 4) At present the disease stays where ever the person was and then decays. Explain how you might model a disease that could spread in the air? (6 marks)

Question-3 (Termites)**[25 marks]**

Answer the following questions by referring to the NetLogo code given below.

- 1) State any assumption made in the modeling of termites gathering wood chips into piles (3 marks)
- 2) What will you observe if you run this program for a considerable time period? (3 marks)
- 3) Describe the setup procedure (5 marks)
- 4) Describe search-for-chip turtle procedure (6 marks)
- 5) Explain the model rules by analyzing the code. (8 marks)

```

to setup
  clear-all
  set-default-shape turtles "bug"
  ask patches
  [ if random-float 100 < density
    [ set pcolor yellow ] ]
  create-turtles number [
    set color white
    setxy random-xcor random-ycor
    set size 5
  ]
end

to go
  search-for-chip
  find-new-pile
  put-down-chip
end

to search-for-chip
  ifelse pcolor = yellow
  [ set pcolor black
    set color orange
    fd 20 ]
  [ wiggle
    search-for-chip ]
end

```

```

to find-new-pile
  if pcolor != yellow
  [ wiggle
    find-new-pile ]
end

to put-down-chip
  ifelse pcolor = black
  [ set pcolor yellow
    set color white
    get-away ]
  [ rt random 360
    fd 1
    put-down-chip ]
end

to get-away
  rt random 360
  fd 20
  if pcolor != black
  [ get-away ]
end

to wiggle
  fd 1
  rt random 50
  lt random 50
end

```

Question-4 (Forest Fire)

[35 marks]

- 1) Describe the model rules for a simple model of forest fire. (Model choice: Assume that the left most edge of NetLogo world is on fire) (10 marks)
- 2) What is the main limitation of the spreading of fire model mentioned above? (4 marks)
- 3) What are the additional factors considered in the below model other than deterministic spread of fire? (Refer to the code given below) (6 marks)
- 4) Analyze the given code and explain it in a detailed manner with an introduction to the model. Note that south-wind-speed refers to the speed of wind coming from South and west-wind-speed refers to the speed of wind coming from the West. (15 marks)

```
globals [
  initial-trees
]

to setup
  clear-all
  ask patches [
    if (random 100) < density
      [ set pcolor green ]
    if pxcor = min-pxcor
      [ set pcolor red ]
  ]
  set initial-trees count patches with [pcolor = green]
  reset-ticks
end
```

*please turn to the next page.

```

to go
  if all? patches [pcolor != red]
    [ stop ]
  ask patches with [pcolor = red] [
    ask neighbors4 with [pcolor = green] [
      let probability probability-of-spread
      let direction towards myself

      if (direction = 0 ) [
        set probability probability - south-wind-speed ]
      if (direction = 90 ) [
        set probability probability - west-wind-speed ]
      if (direction = 180 ) [
        set probability probability + south-wind-speed ]
      if (direction = 270 ) [
        set probability probability + west-wind-speed ]

      if random 100 < probability [
        set pcolor red
        if big-jumps? [
          let target patch-at (west-wind-speed / 5) (south-wind-speed / 5)
          if target != nobody and [ pcolor ] of target = green [
            ask target [
              set pcolor red
            ]
          ]
        ]
      ]
    ]
  ]
  set pcolor red - 3.5
]
tick
end

```

*** End of the Question Paper ***